

Xinrui Wu

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Education

University of Electronic Science and Technology of China (UESTC) Sep 2023 - Present
B.S. in Software Engineering (Embedded System) (GPA 3.63/4) Si Chuan, China

Research Experience

UESTC, CFM (Center for Future Media) Jun 2024 – Present
Research Intern, Advisor: Prof. Yang Yang UESTC, Sichuan

- Research focus: advancing Open-Vocabulary Object Detection (OVD) and Segmentation (OVS).
- Pioneered innovative methods to enhance object detection capabilities, and reduce false positives in OVD.
- Collaborated on developing a robust segmentation framework, improving accuracy across various datasets.
- Achieved significant progress in recognizing and segmenting objects with diverse appearances in images.

Megvii Technology Limited Feb 2025 – Jul 2025
CV Algorithm Intern, Mentor: Dr. Wei Ge MEGVII 旷视 MEGVII, Beijing

- Proposed unified prompt representation method, transforming multi-modal inputs into visual masks to simplify alignment and improve training efficiency.
- Built large-scale automated data pipeline capable of processing billions of samples to support model training.
- Led development of open-vocabulary detection model achieving performance comparable to closed-source SOTA models.
- Successfully deployed the developed OVD model in business applications, validating its practical utility.

HKUST, VGD Group Jul 2025 – Aug 2025
Research Intern, Advisor: Prof. Sai-Kit Yeung HKUST, Hongkong

- Research focus: lightweight image segmentation and salient individual identification for marine environments.
- Proposed the MaskGuide mask-guided distillation framework, which enhances critical information and suppresses noise through a feature separation and purification process.
- The resulting Tiny-MSAM model achieves 99.4% of the Segment Anything Model's accuracy with only 0.5% of its parameters and an inference speed of 164.7 FPS.
- Submitted as a first-author paper to IEEE Robotics and Automation Letters.

Tsinghua, AIR | Dean: Prof. Ya-qin Zhang Jan 2025 – Present
Research Intern, Advisor: Prof. Yuanchun Li, Prof. Yunxin Liu THU, Beijing

- Research focus: generation of controllable length Chain-of-Thought (CoT) for large language models.
- Proposed the BudgetThinker framework, centered on an innovative proportional control token mechanism.
- Via a two-stage training pipeline integrating specialized data and reinforcement learning, our model achieved SOTA on benchmarks like MATH with over 95% budget adherence success, and demonstrated generalization in embodied spatial reasoning.
- Accepted by EMNLP 2026 as a co-first-author paper.

MBZUAI | Dean: Prof. Eric Xing Jan 2026 – Present
Visiting Student, Advisor: Prof. Yutong Xie MBZUAI, Abu Dhabi

- Research focus: optimization of clinical reasoning.
- Proposed a framework that integrates complex multi-stage clinical information to enhance structured medical reasoning.
- Explored frontier training paradigms for clinical world models.

Publication

BudgetThinker: Empowering Budget-aware LLM Reasoning with Control Tokens 2026
Hao Wen*, Xinrui Wu*, ..., Yunxin Liu, Ya-Qin Zhang, Yuanchun Li

Accepted by **EMNLP 2026**

MaskGuide: Efficient Distillation for Deployable Lightweight Segmentation in Marine Environments Apr 2026
Xinrui Wu*, Ziqiang Zheng*, Yiwei Chen, Zeyu Ma⁺, Yang Yang, Sai-Kit Yeung

Accepted by **IEEE Robotics and Automation Letters**

Can Image Models Imagine Time? ImageTime: A Novel Benchmark for Probing Visual World Modeling Through Spatiotemporal Consistency 2026

Xinrui Wu, Lichen Huang
Accepted by **WACV 2027**

HiReT: Hierarchical Relational Trajectory Modeling for Longitudinal Clinical Prognosis <i>Xinrui Wu, Wanqi Yang, Min Xu, Yutong Xie</i> <i>Submitted to ICLR 2027</i>	Sep 2026
Predicting the Future Is Not Enough: Toward Action-Aware Clinical World Models <i>Xinrui Wu, Wanqi Yang, Min Xu, Yutong Xie</i> <i>Submitted to ICLR 2027</i>	Sep 2026

Scholarship

UESTC Model Student Scholarship Top 5% <i>Sci-Tech Innovation</i>	Oct 2025
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Prize

National Second Prize <i>C4 National College Student Network Technology Challenge</i>	Aug 2024
National Third Prize <i>China College Student Computer Design Competition</i>	Jun 2025
National Third Prize <i>National College Student Innovation and Entrepreneurship Competition</i>	May 2025
National Third Prize <i>C4 National College Student Network Technology Challenge</i>	Aug 2024
Second Prize of SiChuan Province <i>China College Student Computer Design Competition</i>	Jun 2024

Interests

Clinical AI: Clinical reasoning, longitudinal prognosis, relational trajectory modeling
World Models: Visual world modeling, spatiotemporal consistency, causal reasoning
Efficient Reasoning & Perception: Budget-aware LLM reasoning, lightweight segmentation, distillation
Spatial Intelligence: 3D perception, vision-language spatial reasoning, AR/VR
Generative Models: Controllable image generation, reference-guided editing, generative model evaluation